Deep Learning Super-Resolution Mapping

Sentinel-2 10 m → 2.5 m with per-pixel trust — pipeline documentation.

Team Sutram · Smart India Hackathon · Problem statement: *Deep Learning Based Super Resolution Mapping (SRM) from Medium Resolution Satellite Imageries* · Document generated 24 August 2026

1. Problem and approach

Medium-resolution imagery (10–30 m) gives broad coverage and frequent revisit, but cannot resolve narrow roads, small buildings, field boundaries or localised damage. The problem statement asks for generative enhancement to <4 m while preserving geographic and spectral consistency.

The central tension

The statement asks for two things that pull against each other: *reconstruct fine-scale detail*, and *clearly manage uncertainty because some reconstructed details are inferred by the model and not directly observed*. A system that only sharpens ignores the second; a system that only stays conservative ignores the first. Our design resolves this by running both kinds of model side by side and never shipping a super-resolved pixel without the map that says how much to trust it.

2. Pipeline architecture

```
Sentinel-2 L2A  (B04 B03 B02 B08 @ 10 m, Copernicus Data Space)
       |
[1] Preprocessing ......... SCL cloud mask, reflectance scaling, tiling w/ overlap
       |
[2] Branches (common interface, run in parallel)
       |-- Bicubic ........ baseline control
       |-- SEN2SR ......... fidelity branch, hard radiometric constraint
       |-- LDSR-S2 ........ generative detail + per-pixel sigma (N diffusion samples)
       '-- Ours ........... trained on SEN2VENuS incl. KUDALIAR (Telangana, India)
       |
[3] Trust layer ........... LR-consistency | spectral angle | branch spread | sigma
       |                               -> fused per-pixel confidence in [0,1]
[4] Reference validation .. Wald protocol, consistency check, calibration curve
       |
[5] Product ............... 2.5 m COG GeoTIFF: 4 SR bands + sigma + confidence
       |
[6] Downstream ............ NDVI, building edges, change detection
```

3. Stage detail

3.1 Preprocessing

- **Cloud masking** via the L2A Scene Classification Layer, rejecting classes 0, 1, 3, 8, 9, 10 (no-data, saturated, cloud shadow, cloud medium/high probability, thin cirrus). Clouds are never super-resolved.
- **Reflectance scaling** from int16 DN to 0–1 float (÷10000), clipped to the physical range.

- **Tiling** at 128×128 with 32 px overlap. Tiles are recombined with a Hann window so overlapping predictions cross-fade instead of leaving seam lines.

3.2 Branches

Branch	Model	Role	Uncertainty
Bicubic	—	Control. Every claim is measured against it.	—
SEN2SR	SPAN CNN, 472k params (ESA OpenSR)	Fidelity. Built-in hard constraint forces radiometric consistency, so these pixels are the analysis-grade output.	—
LDSR-S2	Latent diffusion, 169M params (ESA OpenSR)	Detail. Sharpest imagery, offered for visual interpretation.	Per-pixel σ from N stochastic samples
Ours	SPAN CNN fine-tuned on SEN2VENuS	Team-trained model, specialised on Indian terrain with a physically-grounded objective.	via trust layer

All four expose the same `predict(lr) -> Prediction` interface, so the trust layer and evaluation harness treat them identically and a new branch needs no special-casing.

3.3 Trust layer

Four independent signals, fused into one confidence map:

Signal	What it measures	Why it catches invention
LR-consistency	Downsample the SR output through the Sentinel-2 PSF and compare against the true input	**Physical, not learned.** Disagreement means the model produced radiometry the sensor never recorded.
Spectral angle	Per-pixel SAM between input and downsampled SR	Catches colour drift that L1 error hides; brightness-invariant.
Branch spread	Normalised difference between the fidelity and generative branches	Where a conservative regressor and a diffusion model disagree, the detail is model-invented rather than observed.
Sampling σ	Std-dev across N diffusion samples	Pixels the model is certain about barely move between runs; hallucinated structure varies.

The fused confidence ships as a band *inside* the same GeoTIFF as the imagery — a deliberate choice, so a downstream user cannot load the sharpened pixels without also having the map that qualifies them.

3.4 Reference validation

There is no 2.5 m ground truth for an arbitrary Sentinel-2 scene, so the model is validated where truth does exist, using **Wald's protocol**: degrade the reference ×4, super-resolve it back, and score against the original. Degradation uses a Gaussian approximation of the Sentinel-2 MTF, *not* bicubic — a model trained and tested on bicubic degradation learns to invert bicubic and then fails on real imagery whose blur comes from the instrument optics.

Separately, the **LR-consistency check** requires no ground truth at all and therefore runs on every product we ship.

4. Training data

Branch 4 is trained on **SEN2VENuS v2.0.0** — real Sentinel-2 / VENuS pairs acquired the same day over the same ground, already co-registered.

Why same-day pairing matters

Puri & Kotze (AGILE 2022) traced invented objects in their SRGAN outputs to a 25–30 day acquisition gap between low- and high-resolution imagery: the model learned to reconstruct *change* as if it were *detail*. Same-day pairs remove that failure mode at source rather than patching it later.

Of the 29 sites (139.5 GB total) we take a deliberately small subset:

Site	Location	Patches	Why
KUDALIAR	Telangana, **India**	7,269	Puts real Indian terrain in the training set — the point of the whole exercise for an SIH jury.
ANJI	China	2,312	Mixed agriculture / forest
MAD-AMBO	Madagascar	1,442	Tropical
SUDOUE-4	France	935	Agricultural parcels
ESTUAMAR	Spain	911	Coastal / estuary

4.1 The scale subtlety — and how we resolve it

SEN2VENuS is natively ×2, not ×4

Pairs are 10 m Sentinel-2 against 5 m VENuS (128 → 256 px). Our deployment target is ×4 (10 m → 2.5 m), and no open global 2.5 m reference exists. Stating this openly is stronger than letting a reviewer discover it.

Mode	Input	Target	Scale
`--scale 2`	real S2 10 m	real VENuS 5 m	×2, fully real
`--scale 4`	S2 degraded to 20 m *via the S2 PSF*	**real VENuS 5 m**	×4, real HR target

In ×4 mode only the *input* is synthetic, and it is synthesised with the sensor's own response rather than bicubic. The model therefore learns to invert a realistic sensor degradation against a genuine high-resolution reference. This is Wald's protocol applied to *training* rather than evaluation, and it is what makes a ×4 claim defensible without owning 2.5 m imagery.

4.2 Patch quality filtering

Patches are gated on **contrast**, not on the fraction of zero-valued pixels. These archives carry no no-data sentinel, and roughly 30% of pixels are legitimately zero over dark targets such as open water and closed rainforest canopy. An early zero-fraction filter rejected 129 of 129 patches — it was measuring darkness and discarding whole biomes.

5. Training objective

Deliberately *not* the standard ESRGAN recipe. VGG-19 perceptual loss is trained on 8-bit consumer photographs and is mismatched to 16-bit reflectance; plain adversarial training cost Puri & Kotze roughly 0.2 SSIM in artefacts. We instead optimise what the problem statement actually asks for:

Term	Weight	Purpose
L1 on reflectance	1.0	Radiometric accuracy
LR-consistency through the PSF	0.5	Do not invent radiometry
Spectral angle (SAM)	0.1	Spectral consistency, brightness-invariant
Gradient	0.1	Sharpness without a photographic prior

The consistency term is the interesting one

It is the differentiable form of the check the trust layer runs at inference. The model is penalised *during training* for exactly the behaviour we flag at *deployment* — the objective and the safety check measure the same quantity.

Augmentation is dihedral only (rotations and flips). No colour jitter: shifting reflectance would teach the model that radiometry is negotiable, which is the one thing this pipeline must not learn.

6. Results

6.1 Wald protocol — 512×512 reference tile

Branch	PSNR (dB)	SSIM	SAM (°)	ERGAS	Time
Bicubic	32.68	0.7965	2.025	3.735	1 ms
SEN2SR	**33.21**	**0.8194**	**1.904**	**3.500**	40 ms

For reference, Puri & Kotze (2022) report PSNR ≈ 29.9 / SSIM ≈ 0.71 for SRResNet on Sentinel-2 → 2.5 m near Bangalore. Their figures come from different imagery and are not directly comparable, but they establish the order of magnitude a published baseline achieves on this task.

6.2 LR-consistency on real input — no ground truth required

Branch	Consistency MAE	Consistency SAM (°)	Interpretation
SEN2SR	**0.00384**	**0.880**	Hard constraint doing its job
Bicubic	0.00488	1.032	No invention, but no detail either
LDSR-S2	**0.00586**	**1.577**	Sharpest imagery, furthest from what the sensor measured

This ordering is the empirical justification for the two-branch design. The generative branch produces the most convincing pictures *and* drifts furthest from the observation — so it is offered for visual interpretation while SEN2SR carries the analysis-grade pixels.

6.3 The trust layer is calibrated, not decorative

Binning pixels by predicted confidence and measuring their actual error against ground truth:

Confidence bin	0.0–0.1	0.3–0.4	0.6–0.7	0.8–0.9	0.9–1.0
Mean absolute error	0.0366	0.0296	0.0223	0.0152	**0.0087**

Error falls monotonically as confidence rises — a **4× reduction** from the least to the most confident bin, correlation −0.38. The confidence map genuinely predicts where the model is wrong. Independently, it concentrates its flags on small clustered buildings and road/shadow boundaries: precisely the failure mode Puri & Kotze identified for this class of model, detected without being told to look there.

7. Output product

A Cloud-Optimised GeoTIFF at 2.5 m with six bands:

Band	Name	Contents
1–4	B04, B03, B02, B08	Super-resolved reflectance (0–1)
5	sigma	Per-pixel sampling uncertainty
6	confidence	Fused trust score in [0, 1]

Footprint preservation is verified, not assumed. The affine transform is rescaled so pixel size divides by 4 while the bounding box stays fixed; `check_footprint()` runs after every write and confirms CRS match and bound drift. Measured drift on all products to date: **0.00 m**.

8. Repository and usage

Path	Contents
`src/srm/preprocess/`	SCL masking, tiling, Hann feathering
`src/srm/models/`	Branch wrappers behind one interface
`src/srm/trust/`	LR-consistency, SAM, ΔNDVI, confidence fusion
`src/srm/validate/`	Wald protocol, consistency check, calibration
`src/srm/train/`	Losses, shard dataset, model builder
`src/srm/io/`	COG writing, CRS / footprint preservation
`scripts/`	Fetch, build, train, infer, evaluate
`app/demo.py`	Streamlit demo (runs offline on precomputed products)
`notebooks/`	Colab training notebook, Drive-checkpointed

```
# Super-resolve a scene -> COG with sigma + confidence
python scripts/run_inference.py --input scene.tif --out out/scene_sr \
       --branches bicubic,sen2sr,ldsr,ours

# Benchmark every branch with Wald's protocol
python scripts/evaluate.py --input data/raw/test_ref_2p5m.tif

# Train branch 4
python scripts/fetch_sen2venus.py --sites KUDALIAR
python scripts/build_dataset.py  --sites KUDALIAR --scale 4
python scripts/train.py --data data/interim/sen2venus_x4 --epochs 40 --device mps
```

9. Engineering notes

- **Reflectance-aware throughout.** Metrics use max reflectance = 1.0, not 255.
- **Custom tiling.** `sen2sr.predict_large` derives output height from input *width*, silently corrupting non-square rasters; we replace it with Hann-feathered blending.
- **Streaming dataset build.** Shards are flushed incrementally — buffering KUDALIAR whole is ~4 GB, too close to a free Colab VM's ceiling.
- **Disconnect-tolerant training.** Checkpoints every epoch; `--resume` restores model, optimiser, scaler and epoch counter.
- **Local training is viable.** On Apple Silicon (M5 Pro, MPS) a training step runs ~23× faster than CPU, putting a 40-epoch run at roughly 16 minutes.

10. Honest limitations

- ×4 super-resolution is ill-posed: the output has 16× more unknowns than the input has measurements. Detail below roughly 4 m is *inferred*, not observed.
- We do not claim to reveal objects absent from the input. The claim is sharper delineation of features already partially resolved — and an explicit map of where even that is uncertain.
- The ×4 operator is trained at 20 m → 5 m and applied at 10 m → 2.5 m, which assumes scale-invariance of the degradation.
- Validation against Indian high-resolution references (Bhoonidhi LISS-IV, 5.8 m) is planned but not yet complete.

11. References

1. Aybar et al. *A radiometrically and spatially consistent super-resolution framework for Sentinel-2.* Remote Sensing of Environment, 2025. (SEN2SR)
2. Donike et al. *Trustworthy Super-Resolution of Multispectral Sentinel-2 Imagery with Latent Diffusion.* IEEE JSTARS, 2025. (LDSR-S2)
3. Michel et al. *SEN2VENuS, a Dataset for the Training of Sentinel-2 Super-Resolution Algorithms.* Data, 2022. Zenodo record 14603764, v2.0.0.
4. Puri & Kotze. *Evaluation of SRGAN Algorithm for Superresolution of Satellite Imagery on Different Sensors.* AGILE GIScience Series 3:57, 2022.
5. Wald, Ranchin & Mangolini. *Fusion of satellite images of different spatial resolutions.* PE&RS, 1997. (Validation protocol)
6. ESA OpenSR project — `opensr-test` benchmark framework.